Ahmedabad University

Weekly Report 3

Disease Prediction System using Machine Learning

Submitted to faculty: Mehul Raval

Student: Ayushi Patel

Enrollment No. : AU2519010

Course: CSE 620 Machine Learning

2025-2026 (Winter Semester)

Week 3 Exploratory Data Analysis

Work Completed

This week, exploratory data analysis (EDA) was conducted to better understand the dataset's structure before proceeding with model development. The analysis focused on the distribution of disease classes within the dataset to determine whether there was a significant class imbalance that could influence the behaviour of the machine learning models.

Visual analysis techniques were also applied to gain deeper insights into the dataset. Data visualisation libraries were used to generate plots that illustrate symptom frequencies and their relationships with different diseases.

Through this exploratory analysis, it was observed that several symptoms occur across multiple diseases, suggesting overlap among disease categories. This overlap highlights an important challenge for the classification models, as they must learn to distinguish diseases that share similar symptom patterns.

Tools / Technologies Used

- Python
- Pandas for data handling
- Matplotlib for visualisation
- Seaborn for correlation and heatmap analysis

Challenges Faced

One of the main observations during this stage was the overlap in symptoms across different diseases. This increases the classification task's complexity, as the model must learn subtle differences in symptom combinations.

Plan for Next Week

- Finalise the feature representation for the model.
- Separate the dataset into feature variables (symptoms) and target labels (diseases).
- Prepare the dataset for model training by splitting it into training and test sets.